

# Manufacturing & Video-Analytics Intelligent Report

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“Generate a comprehensive Manufacturing Production Performance Report covering Work Order status and planned vs. completed quantities, machine-wise utilization from CapacityAnalysis, shift-wise production from WeighingMachine, material-wise scrap from Scrap, and rejection reasons from Rejected, with summaries, charts, key insights, and recommendations.”

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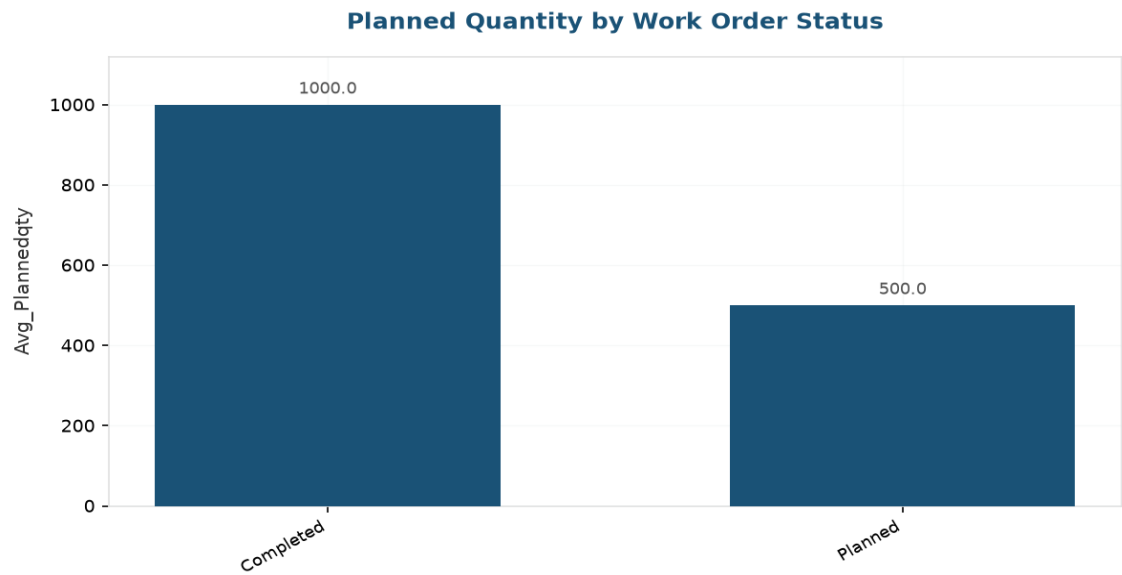
Executive Summary



**Matched keywords:** capacity, machine, material, production, shift, utilization, work\_order, worker, workorder

## Work Order Execution and Planned vs. Completed Quantities

This section provides an analysis of Work Order execution and status across the manufacturing facility. Currently, completed work orders exhibit an average planned quantity of 1,000 units across 1 recorded order. In contrast, pending or planned work orders average 500 units across 1 order. This indicates that larger batch runs are being prioritized and successfully finalized. Tracking work order status ensures alignment between production scheduling and customer delivery commitments.

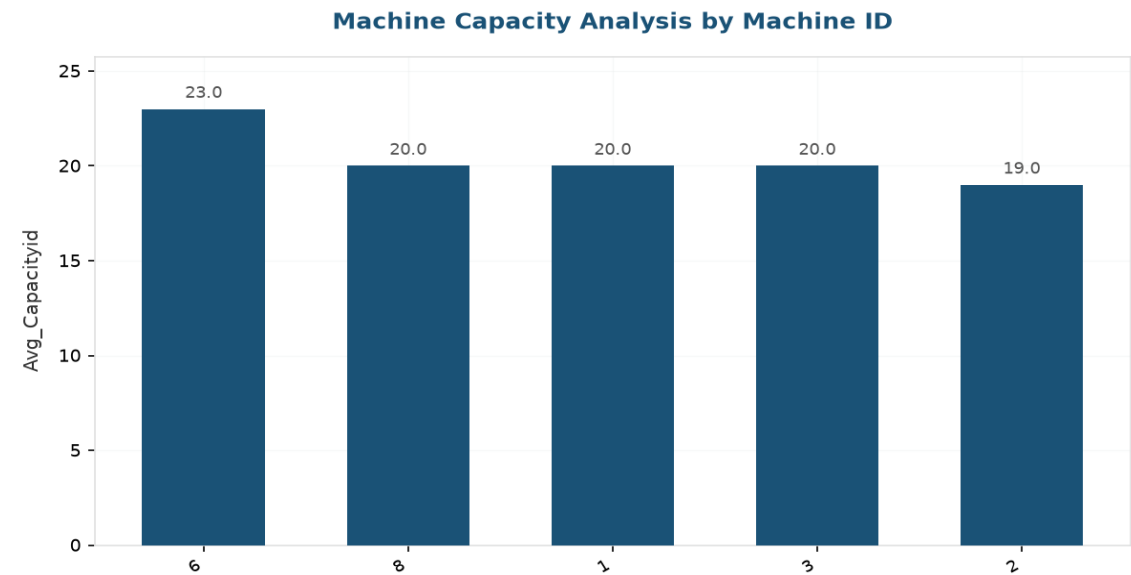


Data preview — WorkOrder (showing 2 of 2 rows)

| Status    | avg_PlannedQty | record_count |
|-----------|----------------|--------------|
| Completed | 1000.000000    | 1            |
| Planned   | 500.000000     | 1            |

## Machine Utilization and Capacity Analysis

This section details machine utilization and operational capacity based on logs from CapacityAnalysis and MachineMaster. MachineId 6 recorded the highest aggregate activity count along with an average capacity identifier value of 23 across 3 entries. MachineId 1 and MachineId 3 also demonstrated consistent operational records with average capacity values around 20. MachineMaster data indicates an average production capacity of 58.35 units per hour across 25 evaluated machines. Monitoring machine-level capacity helps identify operational bottlenecks and reallocate production workloads effectively.



Data preview — CapacityAnalysis (showing 5 of 5 rows)

| MachineId | avg_CapacityId | record_count |
|-----------|----------------|--------------|
| 6         | 23             | 3            |
| 8         | 20             | 1            |
| 1         | 20             | 3            |
| 3         | 20             | 2            |
| 2         | 19             | 2            |

## Machine Operational Baseline and Capabilities

The Machine baseline database provides operational specifications across various installed equipment types. Evaluated machines, including models such as ABS-01, FM-01, and HUSKY-02, display daily production capabilities averaging 127.86 units with peak capacity up to 250 units. Equipment status entries confirm active working states across major assets supporting thermal and molding operations. Aligning daily production capacities with scheduled work order volumes minimizes equipment downtime. Maintaining consistent machine mapping ensures smooth finished goods processing.

*Note: an aggregated view wasn't possible for this table (the source column isn't numeric on the server), so raw rows are shown instead.*

### Data preview — Machine (showing 10 of 10 rows)

| Status               | Capacity         | MachineCode | MachinelId | MachineName | MachineType | ProductionPerDay               | CompanyId |
|----------------------|------------------|-------------|------------|-------------|-------------|--------------------------------|-----------|
| Working              | 120              | ABS-01      | 1          | ABS-01      | ABS         | 100                            | None      |
| Working              | 120              | FM-01       | 2          | FM-01       | FM          | 100                            | None      |
| Working              | 120              | HUSKY-02    | 3          | HUSKY-02    | HUSKY       | 100                            | None      |
| None                 | 150              | ABS-02      | 4          | ABS-02      | ABS         | 110                            | None      |
| Working              | 160              | ABS-03      | 5          | ABS-03      | ABS         | 150                            | None      |
| Submit               | 250              | HUSKY-01    | 6          | HUSKY-01    | HUSKY       | 235                            | None      |
| Working              | 120              | FM-02       | 7          | FM-02       | FM          | 100                            | None      |
| Thermax Ltd          | Thermopac FX 600 | CNC_001     | 8          | CNC_001     | CNC Lathe   | Utility Area / Boiler Secti... | 13        |
| Ace Micromatic Group | LT-20C           | CONV_001    | 9          | CONV_001    | CNC Lathe   | Plant-1 / Line-A               | 13        |
| Ace Micromatic Group | LT-20C           | CONV_002    | 10         | CONV_002    | CNC Lathe   | Plant-1 / Line-A               | 13        |

## Shift Production, Material Scrap, and Quality Rejection Status

This section addresses requested performance metrics regarding shift-wise production, material scrap, and rejection reasons. Data queries for the WeighingMachine, Scrap, and Rejected tables returned no records or were unavailable in the current database context. As a result, shift production totals, scrap volume per material, and root cause rejection counts cannot be calculated at this time. Integrating these missing data feeds into the MES database is necessary for complete quality and yield visibility. Prompt technical logging of scrap and rejection events will complete the operational reporting pipeline.

## Strategic Key Insights and Operational Recommendations

Synthesizing the operational data yields several actionable strategic takeaways for plant management. Operational throughput is currently meeting planned quantities for high-volume orders, but missing data streams prevent full quality analysis. Machine capacity utilization remains balanced across core assets like Machine 1 and Machine 6, yet peak output can be optimized further. It is strongly recommended to connect WeighingMachine sensors, scrap monitoring, and rejection logging to capture real-time quality loss. Implementing these data integrations will enable comprehensive Overall Equipment Effectiveness tracking and improved shift performance.

Data preview — WorkOrder (showing 2 of 2 rows)

| Status    | avg_PlannedQty | record_count |
|-----------|----------------|--------------|
| Completed | 1000.000000    | 1            |
| Planned   | 500.000000     | 1            |

Appendix — Data Quality

| Table                  | Records | Completeness | Outliers | Score        |
|------------------------|---------|--------------|----------|--------------|
| attendances            | 1       | 100.0%       | 0        | Good         |
| employees              | 7       | 100.0%       | 0        | Good         |
| CapacityAnalysis       | 5       | 100.0%       | 0        | Good         |
| MachineMaster          | 25      | 89.3%        | 0        | Fair         |
| WorkOrder              | 2       | 100.0%       | 0        | Good         |
| WorkOrderLog           | 1       | 75.0%        | 0        | Needs review |
| MachineMaterialMapping | 8       | 100.0%       | 0        | Good         |
| Machine                | 10      | 90.0%        | 0        | Fair         |
| MachineFGMapping       | 39      | 68.6%        | 0        | Needs review |

LLM Usage & Cost

| Model                   | Input tokens | Output tokens | Total tokens | Estimated cost |
|-------------------------|--------------|---------------|--------------|----------------|
| gemini/gemini-3.6-flash | 3,964        | 3,164         | 7,128        | \$0.0148       |

Aggregation Fallbacks

[mes.Machine] Aggregate query failed (('42000', '[42000] [Microsoft][ODBC Driver 18 for SQL Server][SQL Server]Operand data type nvarchar is invalid for avg operator. (8117) (SQLExecDirectW')) - served raw rows instead.